Simultaneous Equations & Inequalities

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Linear & Non-Linear Simultaneous Systems**

The diagram shows the graphs of the linear equation $y = 2x + c$ and the quadratic equation $y = -x^2 + 4x + 5$. The two graphs intersect at the points where $x = -1$ and $x = k$. Find the value of c and the value of k.

Final Answer: c = 3, k = 4. The points of intersection satisfy both equations. Therefore, we can set the equations equal to each other: $2x + c = -x^2 + 4x + 5$ Rearranging this equation gives: $x^2 - 2x + (c - 5) = 0$ We are given that the graphs intersect at $x = -1$. Substituting $x = -1$ into the quadratic equation $y = -x^2 + 4x + 5$: $y = -(-1)^2 + 4(-1) + 5 = -1 - 4 + 5 = 0$ So, the point of intersection is $(-1, 0)$. Since this point also lies on the line $y = 2x + c$, we can substitute $x = -1$ and $y = 0$.

2 **[Foundational]** • **Target: Linear and Quadratic Inequalities**

The diagram shows the graphs of two linear equations. The point of intersection of the two lines represents the solution to the system of simultaneous equations. Find the coordinates of the point of intersection.

Final Answer: (2, 1). The point of intersection of two lines is the solution to the system of simultaneous equations that represent these lines. By visually inspecting the graph, we can identify the coordinates where the two lines cross. The first line appears to pass through the points (0, 3) and (3, 0). The second line appears to pass through the points (0, 1) and (2, 3). Let's find the equations of the lines. For the first line: Gradient $m_1 = \frac{0-3}{3-0} = \frac{-3}{3} = -1$. Using the point-slope form wit.

3 **[Foundational]** • **Target: Geometrical Applications & Line-Curve Systems**

Solve the inequality $x^2 + 9x + 20 < 0$.

Final Answer: -5<x<-4. The quadratic factorises as $(x + 4)(x + 5)$ with roots -5 and -4; since the parabola opens upward, the expression is < 0 between the roots, giving $-5 < x < -4$.

4 **[Foundational]** • **Target: Advanced Problem Solving & Exam Challenges**

Solve the inequality $x^2 + 4x + 3 < 0$.

Final Answer: -3<x<-1. The quadratic factorises as $(x + 3)(x + 1)$ with roots -3 and -1; since the parabola opens upward, the expression is < 0 between the roots, giving $-3 < x < -1$.

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5 [Foundational] • Target: Linear & Non-Linear Simultaneous Systems

Solve the inequality $x^2 - x - 6 < 0$.

Final Answer: -2<x<3. The quadratic factorises as $(x - 3)(x + 2)$ with roots -2 and 3; since the parabola opens upward, the expression is < 0 between the roots, giving $-2 < x < 3$.

6 [Foundational] • Target: Linear and Quadratic Inequalities

Solve the inequality $x^2 + 3x - 10 < 0$.

Final Answer: -5<x<2. The quadratic factorises as $(x - 2)(x + 5)$ with roots -5 and 2; since the parabola opens upward, the expression is < 0 between the roots, giving $-5 < x < 2$.

7 [Foundational] • Target: Geometrical Applications & Line-Curve Systems

Solve the inequality $x^2 + x - 20 < 0$.

Final Answer: -5<x<4. The quadratic factorises as $(x - 4)(x + 5)$ with roots -5 and 4; since the parabola opens upward, the expression is < 0 between the roots, giving $-5 < x < 4$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Advanced Problem Solving & Exam Challenges

Solve the inequality $x^2 + 2x - 15 < 0$.

Final Answer: -5<x<3. The quadratic factorises as $(x + 5)(x - 3)$ with roots -5 and 3; since the parabola opens upward, the expression is < 0 between the roots, giving $-5 < x < 3$.

9 [Intermediate] • Target: Linear & Non-Linear Simultaneous Systems

Solve the inequality $x^2 - x - 56 < 0$.

Final Answer: -7<x<8. The quadratic factorises as $(x + 7)(x - 8)$ with roots -7 and 8; since the parabola opens upward, the expression is < 0 between the roots, giving $-7 < x < 8$.

10 [Intermediate] • Target: Linear and Quadratic Inequalities

Solve the inequality $x^2 + 4x - 21 < 0$.

Final Answer: -7<x<3. The quadratic factorises as $(x - 3)(x + 7)$ with roots -7 and 3; since the parabola opens upward, the expression is < 0 between the roots, giving $-7 < x < 3$.

11 [Intermediate] • Target: Geometrical Applications & Line-Curve Systems

Solve the inequality $x^2 + 6x - 7 < 0$.

Final Answer: -7<x<1. The quadratic factorises as $(x - 1)(x + 7)$ with roots -7 and 1; since the parabola opens upward, the expression is < 0 between the roots, giving $-7 < x < 1$.

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12 [Intermediate] • Target: Advanced Problem Solving & Exam Challenges

Solve the inequality $x^2 + 5x - 36 < 0$.

Final Answer: -9<x<4. The quadratic factorises as $(x + 9)(x - 4)$ with roots -9 and 4; since the parabola opens upward, the expression is < 0 between the roots, giving $-9 < x < 4$.

13 [Intermediate] • Target: Linear & Non-Linear Simultaneous Systems

Solve the inequality $x^2 + 3x + 2 > 0$.

Final Answer: $x - 1$. The quadratic factorises as $(x + 1)(x + 2)$ with roots -2 and -1; since the parabola opens upward, the expression is > 0 outside the roots, giving $x - 1$.

14 [Intermediate] • Target: Linear and Quadratic Inequalities

Solve the inequality $x^2 + 7x + 12 > 0$.

Final Answer: $x - 3$. The quadratic factorises as $(x + 4)(x + 3)$ with roots -4 and -3; since the parabola opens upward, the expression is > 0 outside the roots, giving $x - 3$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Geometrical Applications & Line-Curve Systems • Cross-Lesson: 16.01 Permutations & Combinations

Solve the inequality $x^2 + 17x + 72 > 0$.

Final Answer: $x - 8$. The quadratic factorises as $(x + 9)(x + 8)$ with roots -9 and -8; since the parabola opens upward, the expression is > 0 outside the roots, giving $x - 8$.

16 [Challenging] • Target: Advanced Problem Solving & Exam Challenges • Cross-Lesson: 17.01 Discrete Random Variables

Solve the inequality $x^2 - 10x + 21 < 0$.

Final Answer: 3<x<7. The quadratic factorises as $(x - 7)(x - 3)$ with roots 3 and 7; since the parabola opens upward, the expression is < 0 between the roots, giving $3 < x < 7$.

17 [Challenging] • Target: Linear & Non-Linear Simultaneous Systems • Cross-Lesson: 18.01 Binomial Distribution

Solve the inequality $x^2 - 4x - 5 > 0$.

Final Answer: $x5$. The quadratic factorises as $(x + 1)(x - 5)$ with roots -1 and 5; since the parabola opens upward, the expression is > 0 outside the roots, giving $x5$.

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18 [Challenging] • Target: Linear and Quadratic Inequalities • Cross-Lesson: 19.01 Normal Distribution
Solve the inequality $x^2 - 6x - 7 > 0$.

Final Answer: $x7$. The quadratic factorises as $(x + 1)(x - 7)$ with roots -1 and 7; since the parabola opens upward, the expression is > 0 outside the roots, giving $x7$.

19 [Challenging] • Target: Geometrical Applications & Line-Curve Systems • Cross-Lesson: 3.01 Graphs & Transformations
Solve the inequality $x^2 + x - 20 > 0$.

Final Answer: $x4$. The quadratic factorises as $(x - 4)(x + 5)$ with roots -5 and 4; since the parabola opens upward, the expression is > 0 outside the roots, giving $x4$.

20 [Challenging] • Target: Advanced Problem Solving & Exam Challenges • Cross-Lesson: 4.01 Coordinate Geometry
Solve the inequality $x^2 + 7x - 8 > 0$.

Final Answer: $x1$. The quadratic factorises as $(x + 8)(x - 1)$ with roots -8 and 1; since the parabola opens upward, the expression is > 0 outside the roots, giving $x1$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Linear & Non-Linear Simultaneous Systems
Solve the simultaneous equations $y = 2x - 3$ and $x^2 + y^2 = 26$. Which pair of points represents the solutions?

a. $(1, -1)$ **and** $\left(\frac{17}{5}, \frac{19}{5}\right)$ ✓

b. $(3, 3)$ and $(-1, -5)$

c. $(5, 7)$ and $(1, -1)$

d. $(2, 1)$ and $(-2, -7)$

Substitute $y = 2x - 3$: $x^2 + (2x - 3)^2 = 26 \Rightarrow 5x^2 - 12x + 9 = 26 \Rightarrow 5x^2 - 12x - 17 = 0 \Rightarrow (5x - 17)(x + 1) = 0$. So $x = -1 \Rightarrow y = -5$? Wait, check $x = -1$: $(-1)^2 + (-5)^2 = 26$! Yes, $(-1, -5)$ and $(17/5, 19/5)$. Let's verify choice A.

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22 [Foundational] • Target: Linear & Non-Linear Simultaneous Systems

The line $x + y = 5$ intersects the hyperbola $xy = 6$. Find the coordinates of the two intersection points.

a. $(2, 3)$ **and** $(3, 2)$ ✓

b. $(1, 6)$ and $(6, 1)$

c. $(2, 3)$ and $(-2, -3)$

d. $(4, 1)$ and $(1, 4)$

$y = 5 - x \Rightarrow x(5 - x) = 6 \Rightarrow x^2 - 5x + 6 = 0 \Rightarrow (x - 2)(x - 3) = 0$. Points are $(2, 3)$ and $(3, 2)$.

23 [Foundational] • Target: Linear & Non-Linear Simultaneous Systems

Find the points of intersection of the line $2x - y = 1$ and the curve $y = x^2 - 4$.

a. $(3, 5)$ **and** $(-1, -3)$ ✓

b. $(2, 3)$ and $(-2, -5)$

c. $(4, 7)$ and $(1, 1)$

d. $(5, 9)$ and $(0, -1)$

$y = 2x - 1 \Rightarrow x^2 - 4 = 2x - 1 \Rightarrow x^2 - 2x - 3 = 0 \Rightarrow (x - 3)(x + 1) = 0$. For $x = 3, y = 5$; for $x = -1, y = -3$.

24 [Foundational] • Target: Linear and Quadratic Inequalities

Solve the quadratic inequality $x^2 - 2x - 8 < 0$.

a. $-2 < x < 4$ ✓

b. $x < -2$ or $x > 4$

c. $-4 < x < 2$

d. $x < -4$ or $x > 2$

Factor: $(x - 4)(x + 2) < 0$. The parabola is below the x-axis between roots: $-2 < x < 4$.

25 [Foundational] • Target: Linear and Quadratic Inequalities

Find the set of values of x satisfying $2x^2 + 5x - 3 \geq 0$.

a. $x \leq -3$ **or** $x \geq \frac{1}{2}$ ✓

b. $-3 \leq x \leq \frac{1}{2}$

c. $x \leq -\frac{1}{2}$ or $x \geq 3$

d. $-1 \leq x \leq 3$

Factor: $(2x - 1)(x + 3) \geq 0$. Roots at $x = -3$ and $x = 1/2$. Product is non-negative outside roots: $x \leq -3$ or $x \geq 1/2$.

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26 [Foundational] • Target: Linear and Quadratic Inequalities

Solve the simultaneous inequalities $3x - 5 < 7$ and $x^2 - 9 \geq 0$.

a. $x \leq -3$ **or** $3 \leq x < 4$ ✓

b. $-3 \leq x \leq 3$

c. $x < 4$

d. $3 \leq x < 4$

$3x < 12 \Rightarrow x < 4$. $x^2 - 9 \geq 0 \Rightarrow x \leq -3$ or $x \geq 3$. Intersection gives $x \leq -3$ or $3 \leq x < 4$.

27 [Foundational] • Target: Geometrical Applications & Line-Curve Systems

For what range of values of m does the line $y = mx$ not intersect the curve $y = x^2 + 9$?

a. $-6 < m < 6$ ✓

b. $m < -6$ or $m > 6$

c. $-3 < m < 3$

d. $m > 6$

Set $x^2 - mx + 9 = 0$. No intersection requires $\Delta < 0$: $m^2 - 36 < 0 \Rightarrow -6 < m < 6$.

28 [Foundational] • Target: Geometrical Applications & Line-Curve Systems

The perimeter of a rectangle is 28 cm and its area is at least 45 cm^2. What is the range of possible values for the width w ($w \leq l$)?

a. $5 \leq w \leq 7$ ✓

b. $4 \leq w \leq 9$

c. $3 \leq w \leq 7$

d. $w \geq 5$

$l + w = 14 \Rightarrow l = 14 - w$. Area $w(14 - w) \geq 45 \Rightarrow 14w - w^2 \geq 45 \Rightarrow w^2 - 14w + 45 \leq 0 \Rightarrow (w - 5)(w - 9) \leq 0 \Rightarrow 5 \leq w \leq 9$. Since $w \leq l$, width is $5 \leq w \leq 7$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Geometrical Applications & Line-Curve Systems

Find the coordinates of the points where the line $y = 3x - 2$ is normal to the curve $y = x^2 - 4x + 3$.

a. $(1, 0)$ ✓

b. $(2, -1)$

c. $(0, -2)$

d. $(3, 0)$

Curve slope is $dy/dx = 2x - 4$. Normal slope must be $-1/(2x - 4) = 3 \Rightarrow 2x - 4 = -1/3 \Rightarrow 2x = 11/3 \Rightarrow x = 11/6$.

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30 [Intermediate] • Target: Advanced Problem Solving & Exam Challenges

Find the range of values of k for which the line $y = kx - 1$ intersects the circle $x^2 + y^2 = 4$ in two distinct points.

a. $k^2 > -\frac{3}{4}$ **(all real k)** ✓

b. $k > \sqrt{3}$

c. $k < -\sqrt{3}$ or $k > \sqrt{3}$

d. $-1 < k < 1$

Substitute: $x^2 + (kx - 1)^2 = 4 \Rightarrow (1 + k^2)x^2 - 2kx - 3 = 0$.

$\Delta = 4k^2 - 4(1 + k^2)(-3) = 4k^2 + 12(1 + k^2) = 16k^2 + 12$. Since $16k^2 + 12 > 0$ for all real k, the line intersects the circle in two points for all real k.

31 [Intermediate] • Target: Advanced Problem Solving & Exam Challenges

A company finds that its profit P (in thousands of dollars) satisfies $P(x) = -2x^2 + 28x - 80 \geq 10$, where x is the production level in hundreds of units. What is the valid production interval for x?

a. $5 \leq x \leq 9$ ✓

b. $4 \leq x \leq 10$

c. $3 \leq x \leq 11$

d. $6 \leq x \leq 8$

$-2x^2 + 28x - 90 \geq 0 \Rightarrow x^2 - 14x + 45 \leq 0 \Rightarrow (x - 5)(x - 9) \leq 0 \Rightarrow 5 \leq x \leq 9$.

32 [Intermediate] • Target: Advanced Problem Solving & Exam Challenges

Find all values of x satisfying the rational inequality $\frac{x-3}{x+2} < 0$.

a. $-2 < x < 3$ ✓

b. $x < -2$ or $x > 3$

c. $x < 3$

d. $-3 < x < 2$

The quotient changes sign at $x = -2$ and $x = 3$. Testing intervals shows the quotient is negative strictly when $-2 < x < 3$.

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33 [Intermediate] • Target: Linear & Non-Linear Simultaneous Systems

Solve the inequality $x^2 - 25 > 0$.

a. $x6$

b. $x5$ ✓

c. $x \leq -5$ or $x \geq 5$

d. -5<x<5

The quadratic factorises as $(x + 5)(x - 5)$ with roots -5 and 5; since the parabola opens upward, the expression is > 0 outside the roots, giving $x5$.

34 [Intermediate] • Target: Linear and Quadratic Inequalities

Solve the inequality $x^2 - x - 20 > 0$.

a. $x5$ ✓

b. $x \leq -4$ or $x \geq 5$

c. $x4$

d. -4<x<5

The quadratic factorises as $(x + 4)(x - 5)$ with roots -4 and 5; since the parabola opens upward, the expression is > 0 outside the roots, giving $x5$.

35 [Intermediate] • Target: Geometrical Applications & Line-Curve Systems

Solve the inequality $x^2 + 2x - 35 > 0$.

a. $x7$

b. -7<x<5

c. $x5$ ✓

d. $x \leq -7$ or $x \geq 5$

The quadratic factorises as $(x + 7)(x - 5)$ with roots -7 and 5; since the parabola opens upward, the expression is > 0 outside the roots, giving $x5$.

36 [Intermediate] • Target: Advanced Problem Solving & Exam Challenges

Solve the inequality $x^2 - 8x + 12 > 0$.

a. $x \leq 2$ or $x \geq 6$

b. 2<x<6

c. $x6$ ✓

d. $x - 2$

The quadratic factorises as $(x - 6)(x - 2)$ with roots 2 and 6; since the parabola opens upward, the expression is > 0 outside the roots, giving $x6$.

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37 [Intermediate] • Target: Linear & Non-Linear Simultaneous Systems

Solve the inequality $x^2 - 9x + 20 > 0$.

a. $x5$ ✓

b. 4<x<5

c. $x - 4$

d. $x \leq 4$ or $x \geq 5$

The quadratic factorises as $(x - 4)(x - 5)$ with roots 4 and 5; since the parabola opens upward, the expression is > 0 outside the roots, giving $x5$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Linear and Quadratic Inequalities • Cross-Lesson: 19.01 Normal Distribution

Solve the inequality $x^2 - 6x < 0$.

a. **0<x<6** ✓

b. $0 \leq x \leq 6$

c. $x - 6$

d. $x6$

The quadratic factorises as $x(x - 6)$ with roots 0 and 6; since the parabola opens upward, the expression is < 0 between the roots, giving $0 < x < 6$.

39 [Challenging] • Target: Geometrical Applications & Line-Curve Systems • Cross-Lesson: 3.01 Graphs & Transformations

Solve the inequality $x^2 - 3x < 0$.

a. $x - 3$

b. **0<x<3** ✓

c. $0 \leq x \leq 3$

d. $x3$

The quadratic factorises as $(x - 3)x$ with roots 0 and 3; since the parabola opens upward, the expression is < 0 between the roots, giving $0 < x < 3$.

40 [Challenging] • Target: Advanced Problem Solving & Exam Challenges • Cross-Lesson: 4.01 Coordinate Geometry

Solve the inequality $x^2 + 6x > 0$.

a. -6<x<0

b. $x \leq -6$ or $x \geq 0$

c. $x6$

d. $x0$ ✓

The quadratic factorises as $(x + 6)x$ with roots -6 and 0; since the parabola opens upward, the expression is > 0 outside the roots, giving $x0$.

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41 [Challenging] • Target: Linear & Non-Linear Simultaneous Systems • Cross-Lesson: 5.01 Trigonometry

Find the x-coordinates of the points of intersection of $y = x^2 + 9x + 17$ and $y = -3$.

a. x=-5 or x=-5

b. **x=-5 or x=-4 ✓**

c. x=-4 or x=-4

d. x=5 or x=4

Substituting the line into the curve gives $x^2 + 9x + 20 = 0$, which factorises as $(x + 4)(x + 5) = 0$, so $x = -5$ or $x = -4$.

42 [Challenging] • Target: Linear and Quadratic Inequalities • Cross-Lesson: 6.01 Differentiation & Integration

Find the x-coordinates of the points of intersection of $y = x^2 + 10x + 9$ and $y = 2x - 3$.

a. x=-5 or x=-2

b. x=6 or x=2

c. **x=-6 or x=-2 ✓**

d. x=-6 or x=-3

Substituting the line into the curve gives $x^2 + 8x + 12 = 0$, which factorises as $(x + 6)(x + 2) = 0$, so $x = -6$ or $x = -2$.

43 [Challenging] • Target: Geometrical Applications & Line-Curve Systems • Cross-Lesson: 7.01 Statistics & Probability

Find the x-coordinates of the points of intersection of $y = x^2 - 2x - 21$ and $y = -6$.

a. x=-2 or x=5

b. x=3 or x=-5

c. x=-3 or x=4

d. **x=-3 or x=5 ✓**

Substituting the line into the curve gives $x^2 - 2x - 15 = 0$, which factorises as $(x + 3)(x - 5) = 0$, so $x = -3$ or $x = 5$.

44 [Challenging] • Target: Advanced Problem Solving & Exam Challenges • Cross-Lesson: 8.01 Functions: Composite & Inverse

Find the x-coordinates of the points of intersection of $y = x^2 + 4x + 8$ and $y = 5$.

a. x=-2 or x=-1

b. **x=-3 or x=-1 ✓**

c. x=-3 or x=-2

d. x=3 or x=1

Substituting the line into the curve gives $x^2 + 4x + 3 = 0$, which factorises as $(x + 1)(x + 3) = 0$, so $x = -3$ or $x = -1$.

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45 [Challenging] • Target: Linear & Non-Linear Simultaneous Systems • Cross-Lesson: 9.01 Circular Measure

Find the x-coordinates of the points of intersection of $y = x^2 + 6x + 12$ and $y = 4$.

a. x=-3 or x=-2

b. x=-4 or x=-3

c. **x=-4 or x=-2 ✓**

d. x=4 or x=2

Substituting the line into the curve gives $x^2 + 6x + 8 = 0$, which factorises as $(x + 4)(x + 2) = 0$, so $x = -4$ or $x = -2$.

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